LC Resonant Circuit

Final Industrial Electronics Design Project Report

Course: Industrial Electronics | Spring 1403 (2024) | Instructor: Kourosh Khalaj Monfared

Keywords: LC resonant circuit; thyristor; sinusoidal pulse; MOSFET; gate driver; printed circuit board.

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Problem Statement and Input Requirements

The objective of this project is to design an LC resonant circuit that emulates a thyristor-based transcranial magnetic stimulation (TMS) device. The circuit shown below must be designed such that the resulting output current through the inductor takes the form of a sinusoidal pulse with a 200-µs period and a peak current of 60 A, as specified below.

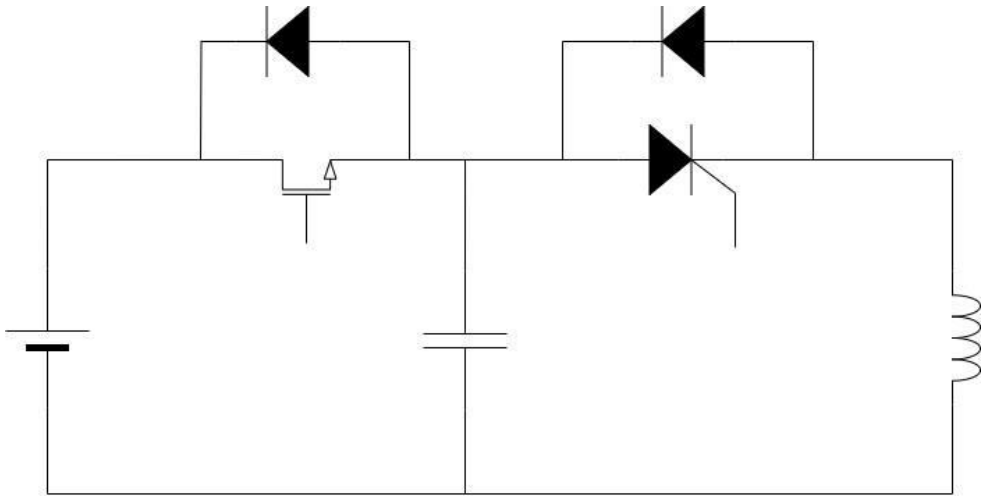


Figure 0-1. LC resonant circuit.

For this purpose, the switching sequence can be implemented as illustrated below.

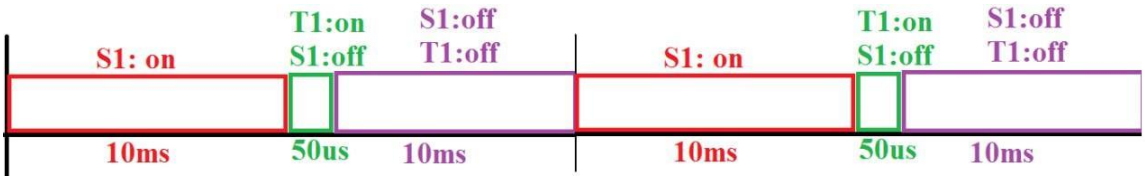


Figure 0-2. Proposed switching sequence.

The intended output waveform is shown below.

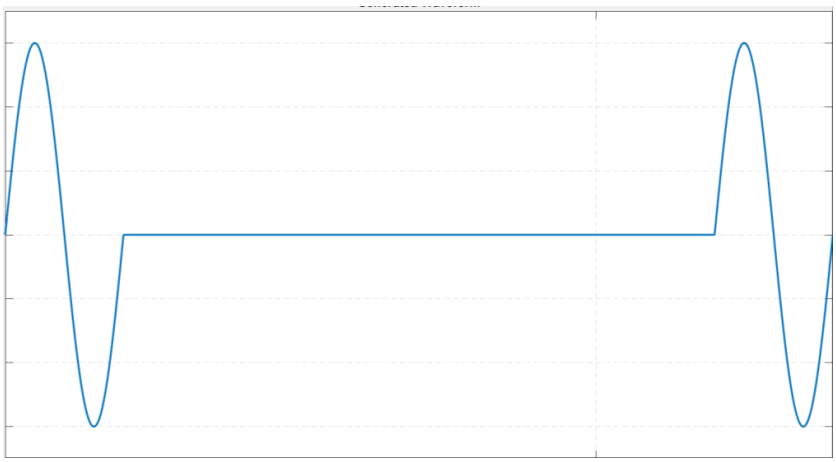


Figure 0-3. Target output waveform.

Part 1 – Design and Selection of Power Components

Output Capacitor Design and Selection

We begin by designing and selecting the required elements, starting with the capacitor and inductor. Since the oscillation period is 200 μs, the following relationship applies:

$$1 / (2\pi\sqrt{LC}) = 1 / (200 \times 10^{-6}) \qquad LC = 1.01 \times 10^{-9}\ \text{H·F}$$

Using this relationship together with Kirchhoff's voltage law (KVL) for the second RLC loop and the impedances of the inductor and capacitor, the following values were obtained:

$$C = 73.4\ \mu\text{F} \qquad L = 13.76\ \mu\text{H}$$

The selected component was a 3.3-μF, 100-V polyester capacitor. Twenty-two such capacitors are connected in parallel to approach the desired capacitance.

Inductor Design and Selection

For this application, the required inductance is approximately 13.7 μH, and an air core must be used. The inductor was manufactured in the laboratory, and its measured value at 10 kHz is presented below.

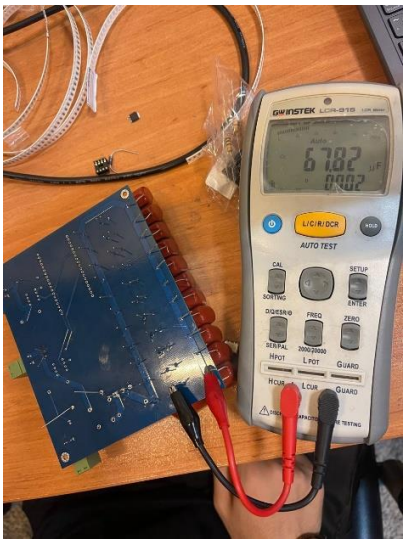


Figure 1-1. Capacitor measurement at 10 kHz.

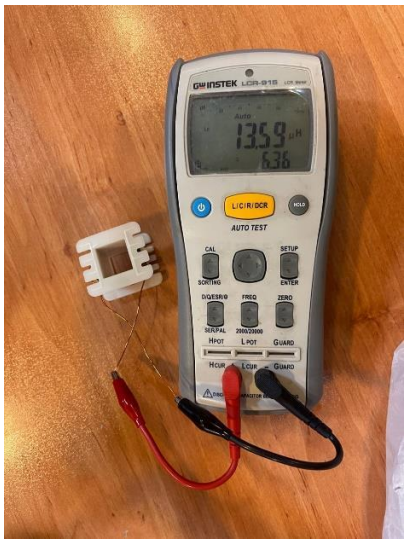


Figure 1-2. Inductor measurement at 10 kHz.

MOSFET Switch Design and Selection

The MOSFET was selected by considering the drain–source (DS) and gate–source (GS) voltages and the current flowing through the switch. Based on these requirements and component availability, the selected device was the 20N06.

Diode Design and Selection

For the diode connected in reverse parallel with the thyristor, the forward current and reverse blocking voltage were considered. The selected component was the SR360 Schottky diode.

Thyristor Selection

The thyristor was selected based on its operating voltage, current, and turn-off interval (tq). Considering these requirements and available components, the BTW68 was selected.

The complete list of required components is presented below.

نام قطعه	تعداد	مشخصات	قیمت	سایت
MBR40100PT	1	100v_40A	79200	https://ickala.com/schottky-diode/25558-mbr40100pt.html
BTW68	1	1200v_30A	278900	https://donya-ic.ir/shop/btw68/
20n06	1	60v_20A	24200	https://ickala.com/mosfet-parts/8709-20n06.html?search_query=20n06&results=1
tlp250	2	Opto	79200	https://ickala.com/optocoupler/21218-tlp250.html?search_query=tlp250&results=1
R3	1	1k	25*374	https://ickala.com/r-dip-5percent-1watt/28563-r1k-1w.html
c1-c11	22	3.3uf , 63v	7700	https://ickala.com/polyster-100v/19505-c33uf100vp.html
R2	2	820ohm	374*25	https://ickala.com/r-dip-5percent-1watt/28562-r820-w1.html
C14	2	100nf	330*25	https://ickala.com/cap-smd-1206/11910-c100nf50vs1206.html?search_query=100nf&results=77
C12	2	100uf	935	https://ickala.com/cap-almin-35v/12452-c100uf35v.html?search_query=100u&results=109
D2	2	zener 20v	5720	https://ickala.com/zener-2w/21194-zy20-20v.html
R4	2	15ohm	25*374	https://ickala.com/r-dip-5percent-1watt/28561-r15-w1.html
R6	2	20k	20*220	https://ickala.com/r-dip-1percent-0-25watt/15524-r20k-w025-1.html
inductor	1	13.5mH	0	ساخت در آزمایشگاه
connector	3	2pin	3762	https://ickala.com/pluggable-terminal/24177-mx126-2pin.html
C	2	100u , 25v	2*35000	https://nikushop.com/tantalum-radial/9806-100uf-25v-tan.html
Rshunt	3	0.01ohm	3*1470	https://www.javanelec.com/#dtl/48736
Rshunt 2	1	0.1ohm	3250	https://www.javanelec.com/#dtl/33565
C100nF25VS1210	2	100nf , 25v	4675	https://ickala.com/capacitor-smd-2413/48100-c100nf-1210.html
Res1	1	10ohm	25*374	https://ickala.com/r-dip-5percent-1watt/16239-r10-1w.html

Figure 1-3. Bill of materials and component sourcing list.

Part 2 – Gate Driver and Feedback Circuit Design

This section presents the design of the gate-driver and feedback circuitry.

Gate Driver Design

An isolated pulse supply is used to control the turn-on timing of the MOSFET and thyristor. A 1-kΩ series resistor and a 10-kΩ parallel resistor are used to limit the current associated with the pulse source. The resulting gate-drive circuit is shown below.

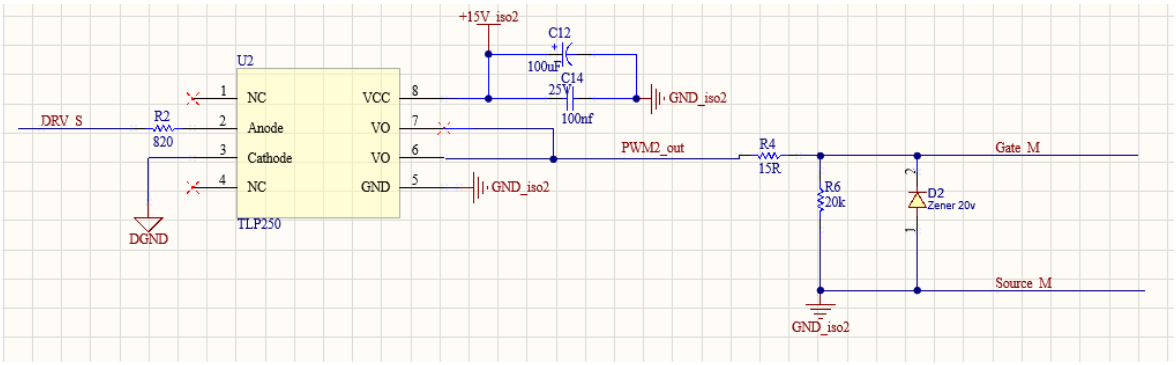


Figure 2-1. Gate-driver circuit.

An optocoupler (photocoupler) is used to control the switching timing and provide electrical isolation between circuit input and output while transmitting the control signal. Such a device is suitable for high-speed switching applications.

Gate Driver Design (continued)

The gate-drive circuit is intended for an IGBT, a power MOSFET, or a thyristor. The TLP250 was selected for this purpose.

In the TLP250, pins 1 and 4 have no physical connection and are unused. Pin 8 supplies the device: the 15-V supply is filtered with 100-nF and 100-µF capacitors to attenuate unwanted frequency components. Pin 5 is ground and provides the return path to the supply. The report specifies a DC supply range of 15–30 V for the TLP250.

The microcontroller signal is applied between pins 2 and 3. To maintain galvanic isolation, the microcontroller ground is connected to pin 3, and the control-signal output is connected to pin 2.

Pins 6 and 7 are internally connected, allowing the output to be taken from either pin. The TLP250 uses a two-transistor totem-pole output arrangement.

A 20-V Zener diode is placed between the output and the supply return to establish the voltage used for controlling the MOSFET and thyristor. A 20-kΩ resistor is connected in parallel with the Zener diode to control the output current.

The voltage across the Zener diode is then applied between the MOSFET gate and source, or between the thyristor gate and cathode, as appropriate. Two copies of the driver circuit are required—one for the MOSFET and one for the thyristor—and their power supplies must be isolated from each other.

Part 3 – Simulation

The final simulation schematic and the waveforms obtained from the simulated circuit are presented below.

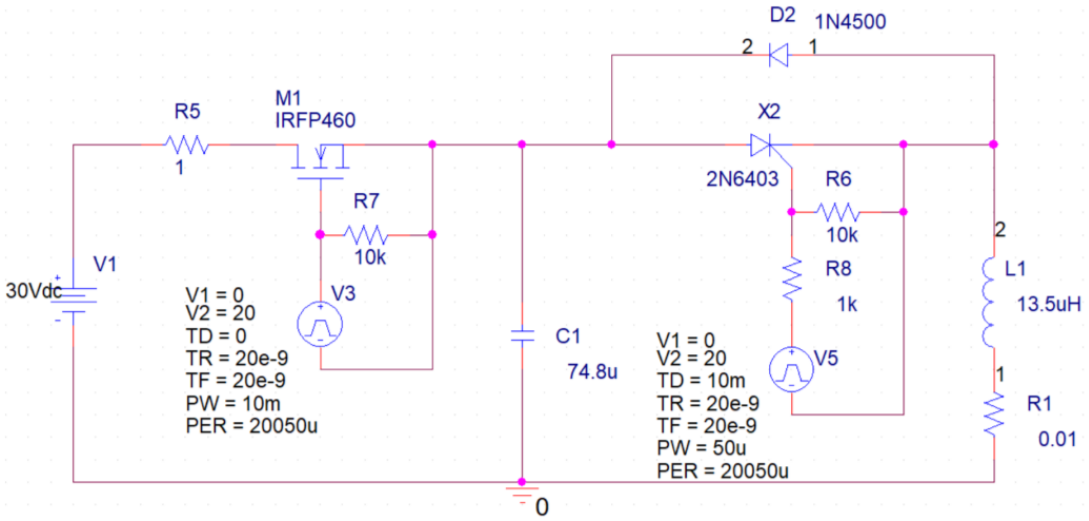


Figure 3-1. Simulated LC resonant circuit.

Part 3 – Simulation Results (continued)

The simulated current through the inductor is shown below.

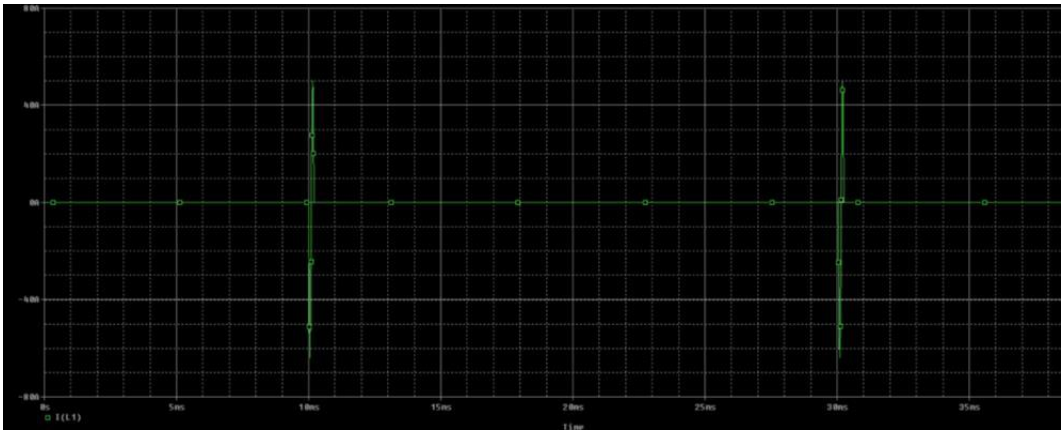


Figure 3-2. Inductor current waveform.

An enlarged view of a single output current pulse is shown below. The waveform reaches the specified peak current and oscillation period.

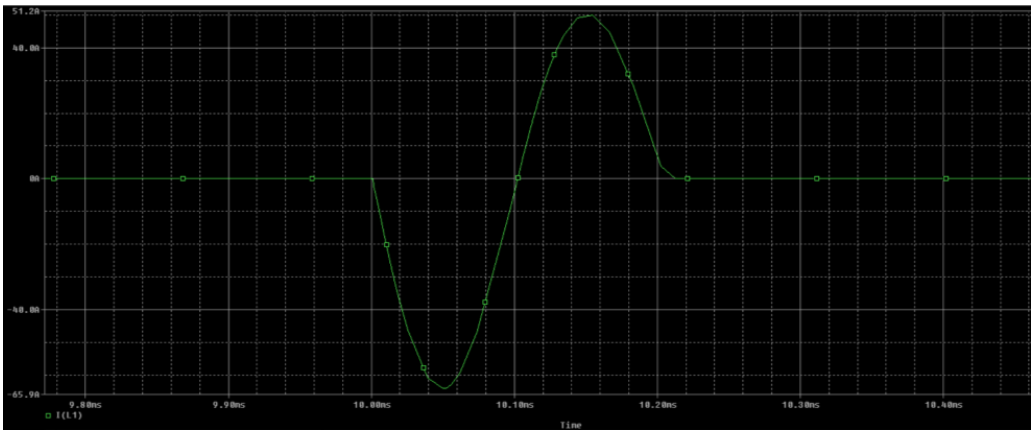


Figure 3-3. Enlarged current pulse.

Part 4 – Controller Programming with a Microcontroller

An ESP32-WROOM 38-pin microcontroller board was used. The controller output was considered to have a voltage of 3.3 V and an available current of 5 mA. Although the board can provide up to 40 mA at its output, the lower value was assumed for additional margin.

```
#include <Arduino.h>

const int MossfetPin = 4; //mossfet pin
const int ThyristorPin = 15; //thyristor pin

void setup() {
    pinMode(MossfetPin, OUTPUT);
    pinMode(ThyristorPin, OUTPUT);
}

void loop() {
    digitalWrite(MossfetPin, HIGH);
    delay(10);
    digitalWrite(MossfetPin, LOW);
    digitalWrite(ThyristorPin, HIGH);
    delayMicroseconds(50);
    digitalWrite(ThyristorPin, LOW);
    delay(10);
}
```

Figure 4-1. Microcontroller program for the switching sequence.

Part 6 – Electronics and Printed Circuit Board (PCB) Design

The intended PCB was designed as shown in the following views, including the circuit schematic, board layout, and a three-dimensional representation of the design.

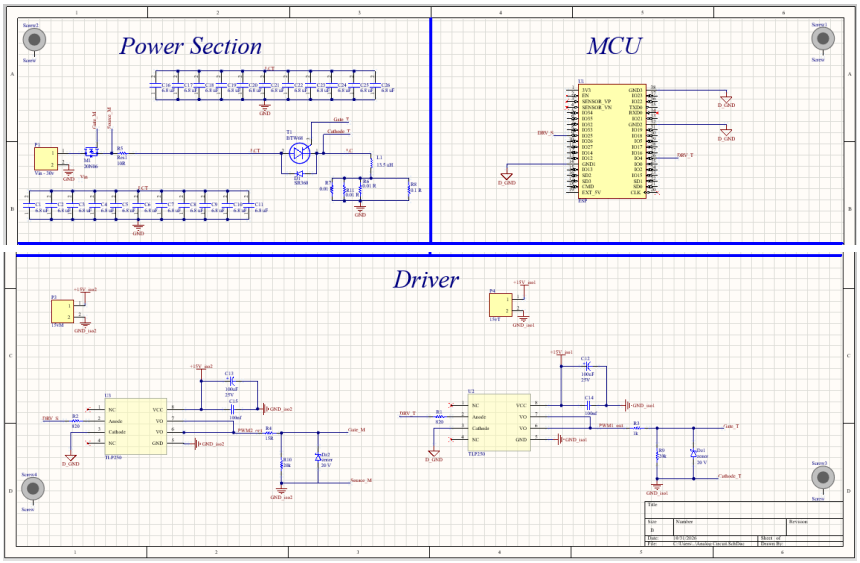


Figure 6-1. Electronic schematic design.

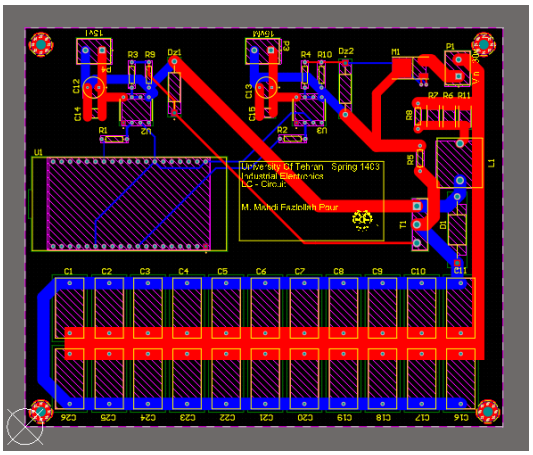


Figure 6-2. PCB layout.

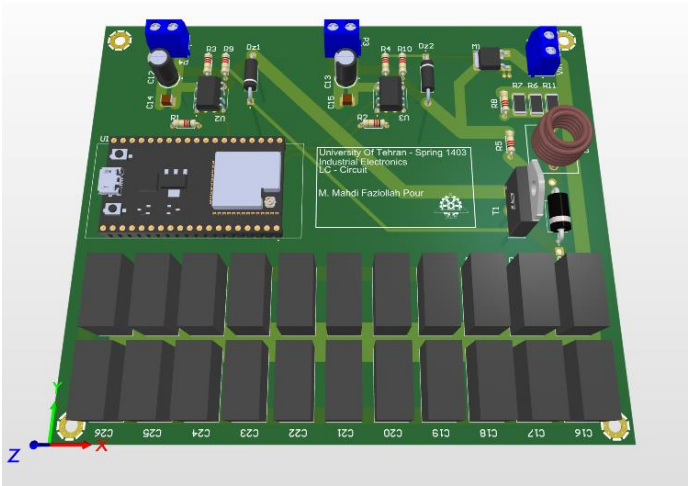


Figure 6-3. Three-dimensional PCB view.

Part 7 – Construction and Testing

After completing the circuit design, procuring the components, obtaining approval for the PCB layout, and manufacturing the board, the circuit was assembled and tested in the Power Electronics Laboratory.

The final implementation and laboratory setup are shown in the following photographs.

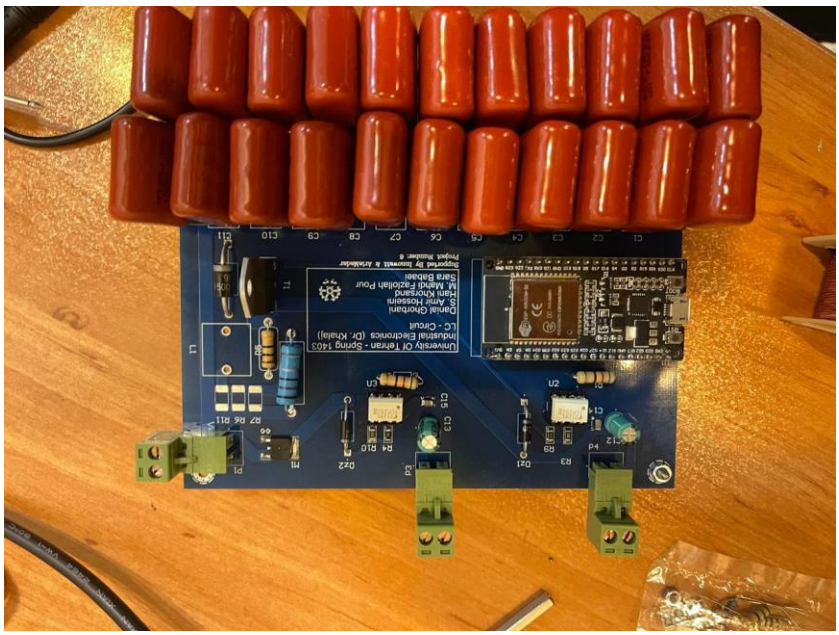


Figure 7-1. Assembled circuit board.

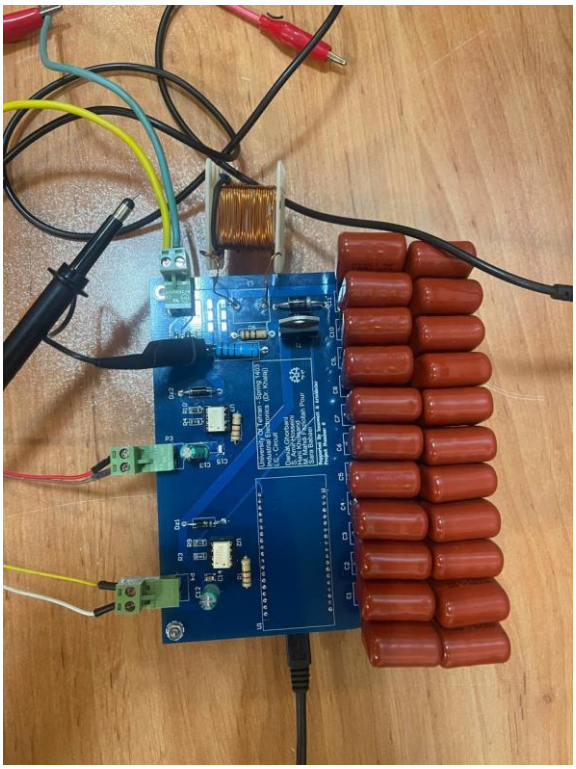


Figure 7-2. Board connected to the four required isolated power supplies.

Part 7 – Construction and Testing (continued)

The measured current waveform through the inductor is shown in the laboratory setup below.

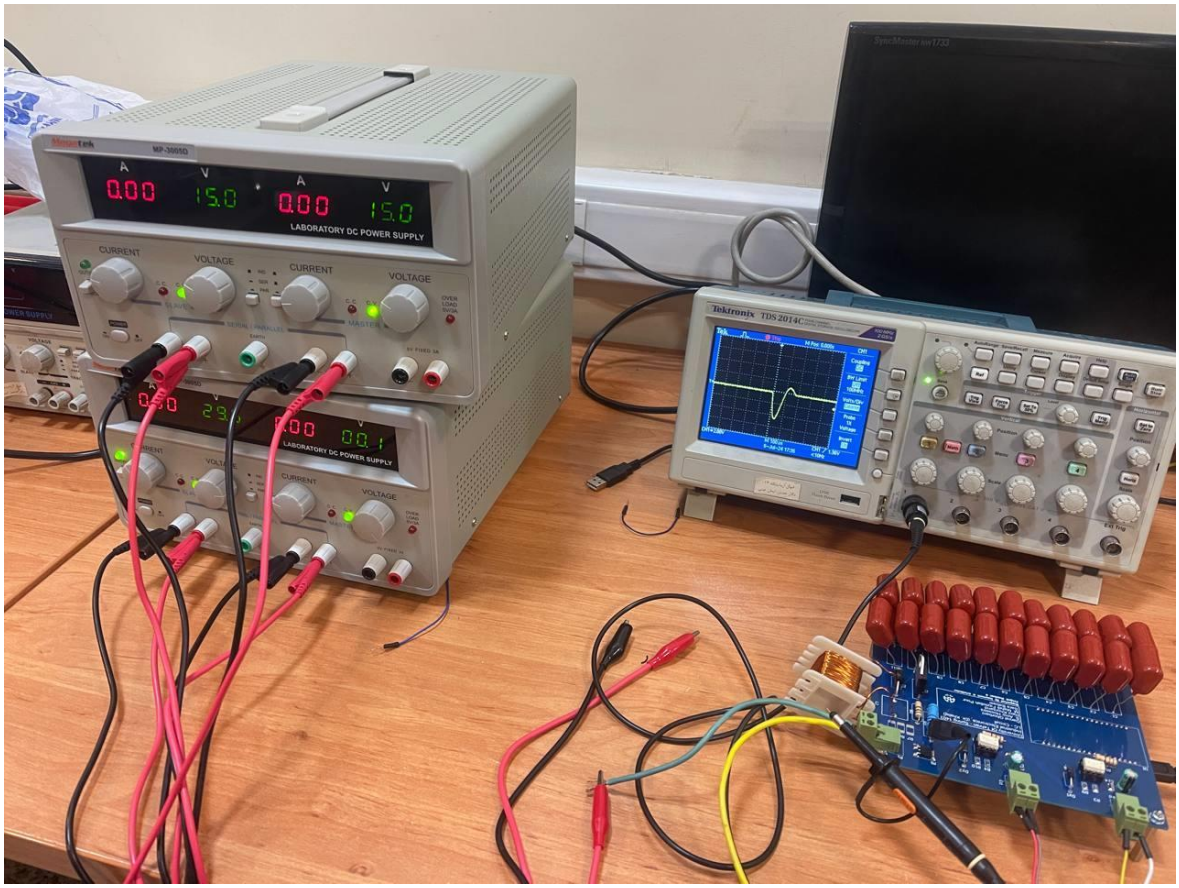


Figure 7-3. Laboratory observation of the inductor current waveform.